EN.530.626: Optimal Control for Space Systems
F2026 Final Project Guidelines

Project Description

As a part of the final project for this course, we encourage student groups of 2 – 3 per team to identify a suitable project topic of mutual interest in the areas of trajectory optimization and optimal control as applied to spacecraft and robotic trajectory planning. We strongly encourage students to formulate and work on this project through the course of the semester with assistance from the teaching team and, as possible, align the final project topic with their current research interests. At the end of the semester, students will present their topic to the class and submit a final project report.

Final Project Proposal

The final project proposal is due on October 13. This project proposal should include details about:

1) A list of team members.
2) The research problem or system of interest that is the focus of the final project.
3) Details on the trajectory optimization, optimal control, or algorithmic tool of interest.
4) An implementation plan (i.e., programming language, interface/solver being used, custom implementation, etc.).
5) A list of possible benchmarks to compare against.
6) Any open questions the team may have on how to accomplish the work.

Final Project Midterm Check-In:

During the week of November 9 – 13, each final project team is expected to meet with the course assistants. This will serve as a project check-in and provide the teaching staff the opportunity to provide feedback and answer any questions students may have.

Final Project Presentations:

Final project presentations will be held in two phases. First, teams will individually meet with the teaching staff on Tuesday December 8 and Thursday December 10, where the teaching staff will review progress and the team's understanding of the work. Thereafter, class-wide final project presentations will be held on Tuesday December 15 during the scheduled final exam slot for the class [1]. Each team will present for approximately ten minutes with two minutes for questions and all team members are expected to speak. Final project presentations will be evaluated for technical clarity and quality and these evaluations will be considered as a part of the final project report grade.

[1] Fall 2026 Examination Schedule

Final Project Report:

The final project report will be due on December 8. The final project This report should be six pages long and written in LaTeXusing the IEEE two-column conference template [2]. The project report should mirror a standard conference publication and include:

1) Introduction section discussing the motivation and problem statement.
2) Relevant work section discussing existing approaches to address this work and pertinent algorithms that have been studied in the plan.
3) Problem formulation section detailing the trajectory generation problem of interest and details on the algorithmic approach used.
4) Results section providing implementation details and a comparison against benchmarks where possible.
5) Conclusion section summarizing the work and identifying possible areas of future improvement and investigation.
6) A list indicating the contributions made by each individual team member on the project.

Example Topics

The instructor team is happy to provide feedback and guidance on suitable topics of interest. A few examples of appropriate project scope include:

1. Developing a GPU-compatible sequential quadratic programming solver using `jax` [1].
2. Integrating large language models for code generation of optimal control solvers [1, 2].
3. Leveraging implicit function theorem for differentiable optimal control [1, 2].

FAQs

1. I have another class in which there is a final project. Can I count the same work towards final projects in these two (or more) classes? No, students are expected to complete a distinct and unique final project as a part of this class.
2. Can I count work from my research essay or PhD. thesis towards the final project? Students are encouraged to use the final project as an opportunity to explore further research topics related to their research essay/PhD. thesis. However, students may not simply reuse previously completed work for this final project.
3. Can I explore ideas from machine learning or reinforcement learning for the final project? Students may certainly incorporate technical knowledge from outside the class as a part of their final project. However, the project topic is still expected to meaningfully connect with topics from optimal control, trajectory optimization, among others discussed in the class.
4. One of my project team members has not contributed towards our group effort. If there are issues with team members not participating or contributing towards the final project, then the remaining students are encouraged to email the teaching staff to raise any issues or concerns. Further, students will be provided an opportunity to list individual contributions as a part of the final project report.

[2] https://www.ieee.org/conferences/publishing/templates